

Customer Churn: Executive Summary

What's driving cancellations, how confident we can be, and what to do next

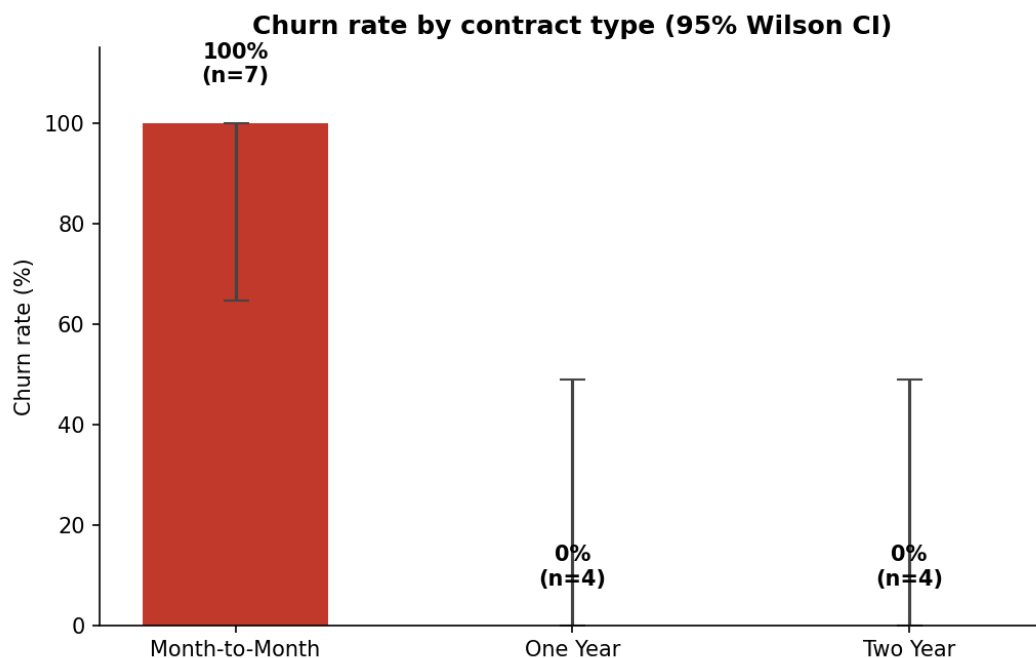
Read this first — sample size caveat. This analysis is based on a dataset of **15 customers** (7 churned, 8 retained). That is enough to spot candidate patterns worth investigating, but far too small to treat any single number below as a precise, stable estimate of company-wide behavior. Ranges shown are 95% confidence intervals — notice how wide most of them are. Treat every finding as a **hypothesis to validate** against the full customer base before committing budget.

46.7%	32.3%	\$409.93	100% vs 0%
Sample churn rate (7 of 15 customers)	Share of monthly revenue held by churned accts	Monthly recurring revenue lost in this sample	Churn rate: month-to-month vs. annual/multi-year contracts

The problem

Nearly half of the customers in this dataset have churned, and those cancellations represent roughly a third of monthly recurring revenue in the sample. Leadership needs to know: is this churn random, or does it cluster around specific, actionable conditions we can change? The analysis below identifies five patterns that consistently separate churned from retained customers in this data, quantifies how strong each pattern is, and translates that into a recommended first action.

Insight 1 — Contract type is the strongest churn signal we found

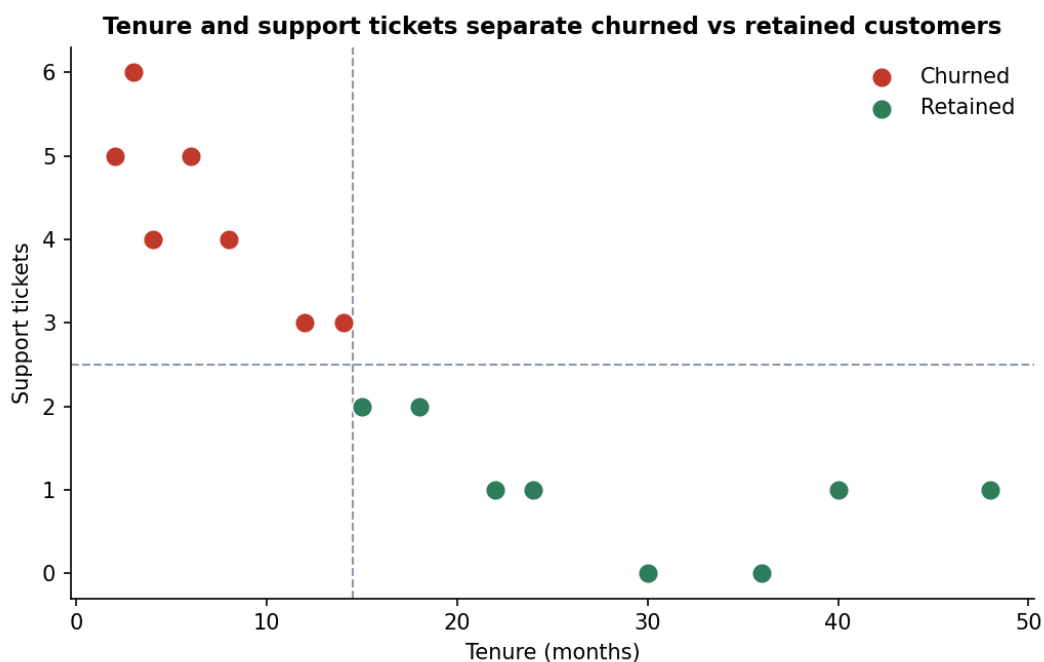


Churn rate by contract type, with 95% confidence intervals (Wilson method) reflecting sample uncertainty.

Every one of the 7 month-to-month customers in this sample churned (100%, 95% CI: 65%–100%); every one of the 8 customers on a one-year or two-year contract stayed (0%, 95% CI: 0%–49% for each group). This is a Fisher's exact test p-value of <0.001 — too clean to be noise, but also too clean for 15 customers, which is a flag rather than a green light.

Assumption/limitation: a perfect 100%/0% split in a dataset this small is more consistent with a strong underlying effect than with a subtle one that only shows up at scale — but with zero counterexamples in either direction, we cannot yet rule out a data-generation artifact (e.g., how this sample was selected). Before resourcing a retention program around this, confirm the same split holds in the full account base, ideally with month-to-month churn checked against a denominator in the hundreds or thousands.

Insight 2 — Support ticket volume separates churned and retained customers almost perfectly



Each dot is one customer. Dashed lines mark the thresholds that separate the two groups in this sample.

Churned customers logged 3–6 support tickets (average 4.3); retained customers logged 0–2 (average 1.0). No customer in this sample crossed both thresholds in the "wrong" direction — the two groups sit on opposite sides of roughly 2–3 tickets. Elevated support contact is a plausible early-warning signal, and one that is easier to instrument (via ticket volume) than contract type is to change.

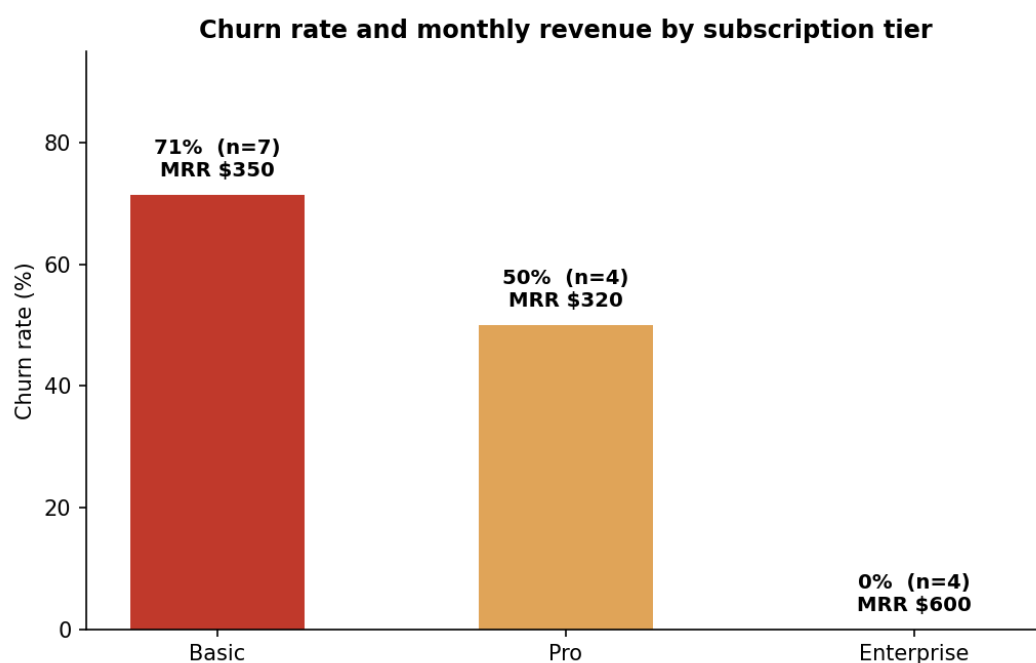
Assumption/limitation: we don't know from this data whether tickets are a cause of dissatisfaction or a symptom of it (customers already leaning to churn may simply complain more before they leave). Treat 3+ tickets as a monitoring trigger, not a proven causal lever, until a cause can be established (e.g., ticket category/resolution time analysis).

Insight 3 — Churn is concentrated in the first 14 months of tenure

Every churned customer in the sample left within 2–14 months of tenure (average 7 months). Every retained customer had 15–48 months of tenure (average 29.1 months). The two groups do not overlap in this sample, pointing to an early-relationship risk window rather than a steady, constant-hazard churn rate over a customer's lifetime.

Assumption/limitation: this pattern is partly mechanical — a customer must survive long enough to accumulate tenure, so "churned early" and "low tenure" are related by definition, not purely by cause. It still has practical value as a targeting rule (concentrate retention outreach in months 1–14), but should not be read as proof that risk drops to zero after month 14; we simply have no customer in this sample who churned later than that to compare against.

Insight 4 — Lower-priced tiers churn more, but the highest-revenue tier hasn't churned at all



Churn rate and total monthly recurring revenue (MRR) by subscription tier.

Basic-tier customers churned at 71% (5 of 7, 95% CI: 36%–92%), Pro at 50% (2 of 4, 95% CI: 15%–85%), and Enterprise at 0% (0 of 4, 95% CI: 0%–49%). Enterprise, the highest-priced tier, holds the most revenue at risk of nothing so far — but it's also the smallest, most confidence-interval-wide group, so "Enterprise never churns" is not yet a safe conclusion.

Assumption/limitation: tier is correlated with contract type and tenure in this sample (Enterprise customers tend to be on longer contracts with more tenure), so this may not be an independent driver — it could be the same underlying effect as Insights 1 and 3 viewed through a different variable. A larger dataset with more independent variation between tier, contract, and tenure is needed to separate these.

Insight 5 — Payment method shows a suggestive but statistically weak split

UPI customers churned at 75% (3 of 4) and debit-card customers at 100% (2 of 2); bank-transfer customers churned at 0% (0 of 3) and credit-card customers at 33% (2 of 6). Directionally, digital/instant payment methods look riskier than bank transfer in this sample.

Assumption/limitation: two of these four groups have only 2–3 customers each, so the 95% confidence intervals span nearly the entire 0–100% range (e.g., debit card: 34%–100%). This is the weakest of the five findings and is included mainly to flag it for monitoring, not to act on directly — it should not drive any payment-method policy change on its own.

At a glance

Insight	Churned group	Comparison group	95% CI width	Confidence
1. Contract type	100% (7/7) MTM	0% (0/8) annual+	Narrow-to-mod.	High
2. Support tickets	avg 4.3 (3–6)	avg 1.0 (0–2)	N/A (clean split)	Moderate-High
3. Tenure	avg 7 mo (2–14)	avg 29 mo (15–48)	N/A (clean split)	Moderate
4. Subscription tier	71% Basic churn	0% Enterprise churn	Wide	Moderate
5. Payment method	75–100% UPI/Debit	0–33% Bank/Credit	Very wide	Low

Recommended action

Step 1 (validate, low cost, ~1–2 weeks): Re-run this same analysis against the full customer base. Confirm whether the month-to-month vs. annual-contract split and the support-ticket threshold hold at scale. This is the single highest-leverage next step because Insight 1 is the strongest pattern found but rests on only 15 customers.

Step 2 (pilot, if confirmed): Target month-to-month customers in months 1–14 of tenure who have logged 3+ support tickets — the highest-overlap-risk group across three of the five insights — with a proactive save offer (e.g., a discounted incentive to move to an annual contract). Measure the pilot's effect on churn rate and revenue retained before rolling out further.

Step 3 (deprioritize for now): Do not build a payment-method-specific intervention (Insight 5) until it can be tested on a larger, more balanced sample — the current evidence is too thin to justify the operational cost.

Bottom line: the data points toward contract structure, early-tenure engagement, and support friction as the most defensible places to intervene — but every number in this report should be read as "worth testing," not "proven," until validated on a larger population.

Prepared from customer_churn_cleaned.csv (n=15). Confidence intervals use the Wilson score method, appropriate for small-sample proportions. This report accompanies an interactive dashboard and a walkthrough presentation.